For better clinical understanding systolic and diastolic pressure measurements may be taken in both arms.

Types of sphygmomanometer

1. **Manual sphygmomanometer**
   - **a. Mercury sphygmomanometer:** It shows blood pressure by affecting the height of a column of mercury which does not require calibration.
   - **b. Aneuroid sphygmomanometer**: It is a mechanical type with a dial and it requires calibration checks unlike mercury manometer.
2. **Digital sphygmomanometer:** It detects systolic and diastolic pressures by an oscillometric detector. It is easy to use even without training.





Figure 12.2 Sphygmomanometer – Mercury and Digital

The maximum pressure exerted during the ventricular contraction is called **systolic pressure** and the minimum pressure exerted during the ventricular relaxation is called **diastolic pressure.**

Normal Blood Pressure	= 120/80 mmHg
Systolic pressure	= 120 mm Hg
Diastolic pressure	= 80 mm Hg



"WHITE COAT EFFECT"

White coat hypertension, more commonly known as **white coat syndrome**, is a phenomenon in which the patients exhibit a blood pressure level above the normal range, in a clinical setting, though they do not exhibit in other settings.

Clinical Significance

1. To diagnose pathological conditions such as hypertension and hypotension
2. Helps to assess the state of blood circulation.
3. Provides the functional details of heart.

Glucometer

Glucometer is a simple and portable medical device used to record the approximate levels of blood glucose. It is a battery operated digital meter.



Figure 12.3 Glucometer

A drop of blood is taken from the tip of the finger and placed on the disposable test

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